INCH-POUND

MIL-C-48882D (AR)
AMENDMENT 3
01 February 2001
SUPERSEDING
AMENDMENT 2
23 August 1994

MILITARY SPECIFICATION

CHARGE, PROPELLING, 81MM
PARTS AND CONTAINER ASSEMBLY

This amendment forms a part of MIL-C-48882D (AR), dated 5 March 1990, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, under Drawings: Change title for Drawing 9313721 as follows:

"Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling or Container Assembly."

PAGE 4

3.9, Delete in its entirety and substitute the following:

"3.9 Homogeneity of material. The material used to manufacture the parts and assemblies shall be uniform in structure or composition throughout. The presence of spots ¼ inch or larger shall indicate non-homogeneity. The presence of thin membrane-like areas in the nose sections of the part or assembly shall indicate non-homogeneity. The top and bottom halves shall not exhibit non-homogeneities including cracks, splits, cuts, tears, rips or holes when tested as specified in 4.5.7. The Government shall establish visual standards prior to production with the assistance of the contractor."

PAGE 7

Table I, Closure: Delete "Composition … 1/ … 100 2/ … 3.2 … 4.5.1" and substitute "Composition … 1/ 4/ … 100 2/ … 3.2 … 4.5.1"

Container Assembly: Delete "Composition … 5 2/ … 3.2 … 4.5.1" and substitute "Composition 4/ … 5 2/ … 3.2 … 4.5.1"

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Notes, add Note 4 as follows: "4/ Microphotographs shall serve as evidence of acrylic fibrillation."

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* 4.4.1a, Delete subparagraphs "(2), (3), (4) and (5) in their entirety and substitute the following:

"(2) Diphenylamine from not more than two lots from one manufacturer.
(3) Acrylic fibers from not more than two lots from one manufacturer.
(4) Polyester fibers from not more than two lot from one manufacturer.
(5) Kraft fibers from not more than two lot from one manufacturer."

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4.2.2.2, Delete Minor 201.

PAGE 12

4.4.2.4, Major 101: Delete "101 … Composition 1/ … 100 2/ … 3.2 … 4.5.1" and substitute "101 … Composition 1/ 4/ … 100 2/ … 3.2 … 4.5.1"

Minor 201, Delete "Length 3/" and substitute " Length (if applicable) 3/"

Add the following:

"Minor	Title	AQL	Reg Para.	Para. Ref
205	Diameter (if applicable) 3/"	0.65%	3.2	Gage

Notes, add Note 4 as follows: "4/ Microphotographs shall serve as evidence of acrylic fibrillation."

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4.4.2.5, Major 106: Delete "106 Composition … 5 1/ … 3.2 … 4.4.3.1/4.5.1" and substitute "106 Composition (3/) … 5 1/ … 3.2 … 4.4.3.1/4.5.1"

Notes, add Note 3 as follows: "3/ Microphotographs shall serve as evidence of acryilic fibrillation."

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4.4.2.6, Title block, Delete "Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling, M205 or Container Assembly" and substitute "Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling or Container Assembly."

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4.4.3.1, Delete in its entirety and substitute the following:

"4.4.3.1 Composition. The container moldings and closure material shall be selected from each lot and tested in accordance with 4.5.1. The container moldings shall be free of silicone. The lot shall be rejected if the sample fails to comply with the requirements."

4.4.3.2, Delete in its entirety and substitute the following:

"4.4.3.2 Stability heat test (134.5 C). The container moldings and closure material shall be selected from each lot and tested in accordance with 4.5.2. The container moldings shall be free of silicone. The lot shall be rejected if the sample fails to comply with the requirements."

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4.5.1.3.3.1, Delete in its entirety and substitute the following:

"4.5.1.3.3.1 Nitrocellulose – Schulze-Tiemann method for nitrocellulose (NC) determinations. Calibrate a eudiometer tube prior to assembly of the Schulze-Tiemann apparatus by suspending the tube in an upside down position and filling this with incremental volumes of aqueous methanol solution from a buret. Take a reading on the eudiometer tube for each volume of aqueous methanol transferred from the buret. Subtract each volume measured with the buret from the corresponding reading on the eudiometer tube. Take the average of the difference between the buret and eudiometer readings. This average is the eudiometer volume correction X.

The value of X is positive if the buret readings are smaller than the eudiometer readings and vice versa.

Assemble a 250 ml round bottom flask, gas evolution apparatus, eudiometer tube, NaOH bath, lead band, heating mantle and leveling bulb as shown in Figure 1.

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Calibrate the apparatus with dry standard A.C.S. grade KNO_3. Follow the procedure of analysis below using 0.5 – 0.7 g of standard KNO_3 weighed to the nearest 0.1 mg. Calculate the factor of the apparatus according to the following equation:

$$F = \frac{616.8W\ (273.15 + T)}{(V\text{-}X)\ (B\text{-}C\text{-}P\text{-}3.48)}$$

Where,
- B = barometric pressure (mm Hg)
- C = temperature correction (mm Hg), See 4.5.1.3.3.2
- F = S-T method correction factor
- P = vapor pressure of water over 33 percent w/w NaOH at T (mm Hg), See 4.5.1.3.3.3
- V = volume of gas in eudiometer tube (ml)
- W = mass of KNO_3 (g)
- X = eudiometer volume correction (ml)

Prepare samples as follows: Cut sample into ¼ inch pieces or smaller. Dry to constant weight using the microwave moisture analyzer, then store in a desiccator.

Fill the eudiometer tube with the NaOH solution by means of the leveling bulb. Weigh between 0.7 and 1.0 g of specimen to the nearest 0.1 mg and transfer this into the round bottom flask. Add 30 ml of deionized (DI) water, heat the water to boiling and purge the flask/gas evolution apparatus assembly with steam for 5 to 10 minutes. Remove the heat for a minute, then add 25 ml of $FeC1_2$ solution, 25 ml concentrated HC1, and 10 ml of DI water.

Place the eudiometer tube over the tip of the gas evolution apparatus and allow the reaction to continue for 20-30 minutes or until the bubbling has stopped. Add 30 ml of DI water through the addition funnel in a similar manner and continue heating for 5 to 10 minutes more. Move the eudiometer away from the gas evolution apparatus, turn off the heat, and release the vacuum in the flask by opening the addition funnel stopcock. Allow the eudiometer to stand for at least 10 minutes, then read the gas volume on the eudiometer to the nearest 0.1 ml at equilibrium pressure with the atmosphere using the leveling bulb.

Calculate the NC content of sample according to the following equation:

$$\text{Percent NC} = \frac{2.246F\ (V\text{-}X)\ (B\text{-}C\text{-}P\text{-}3.48)}{NW\ (273.15+T)}$$

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Where, B = barometric pressure (mm Hg)
C = temperature correction (MM Hg), See 4.5.1.3.3.2
F = S-T method correction factor
N = percent nitrogen in NC
P = vapor pressure of water over 33 percent w/w NaOH at T (mm Hg), See 4.5.1.3.3.3
V = volume of gas in eudiometer tube (ml)
W = mass of sample (g)
X = eudiometer volume correction (ml)

Preparation of $FeC1_2$ solution. Prepare the solution by adding 1000 g of $FeC1_2.4H_2O$, 1200 ml DI water, and 70 ml of concentrated HC1 into a 5 liter round bottom flask. Attach a reflux condenser and reflux for 20 minutes. Cool to room temperature. Slowly add 20 grams of iron filings or powder to the cool solution. When the gas evolution stops, filter quickly. Use A.C.S. grade chemicals.

Preparation of NaOH solution, 33 percent (w/w). Dissolve 330 g of reagent grade NaOH in 670 g of DI water.

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Delete Figure 1 and substitute a new Figure 1 as follows:

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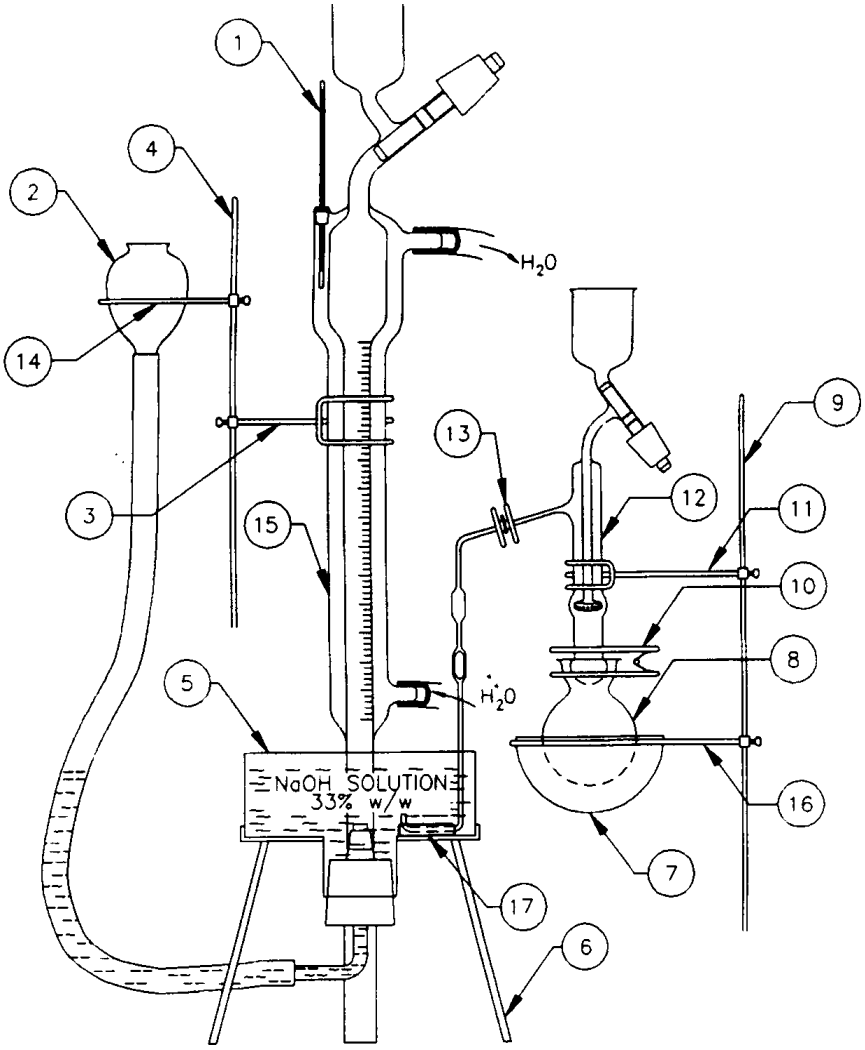
Laboratory setup for the determination of nitrocellulose
by the Schulze-Tiemann method



Schulze-Tiemann Apparatus

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4.5.1.3.4.1, Delete in its entirety and substitute the following:

"4.5.1.3.4.1 Alternate method – preparation of samples. Each sample shall be cut into approximately ¼ inch squares or circles. Cut and dry, in microwave or oven, approximately 2.5 g from each sample and cool in desiccator. From each sample, approximately 1.0 g test specimen shall be weighed to the nearest 0.2 mg. This test is destructive and the sample shall be removed from the lot.

Analytical procedure

Solvent extraction. Place the test specimen in a centrifuge bottle and add about 50 ml of absolute methanol. Completely homogenize the specimen in methanol for approximately 2 minutes with a polytron homogenizer. Turn the homogenizer to low speed, then with a squirt bottle of technical grade acetone, slowly add the acetone down the shaft of the homogenizer until the level reaches about ½ inch below the shoulder of the bottle. Stir at slow speed for 2 minutes. Centrifuge the contents in the bottle for 5 minutes and decant the liquid. Repeat the dispersion of the specimen in acetone without methanol, centrifuge the mixture, and decant the clear liquid. Rinse the polytron shaft with acetone and save the rinsing in the centrifuge bottle containing the specimen. Set up a vacuum filter equipped with an appropriate filter paper. Swirl the contents in the bottle, then pour the entire mixture into the filter funnel. Turn on the vacuum. Rinse the shaft with more acetone, catching the rinsing in the bottle, and pouring this into the filter funnel until all the fibers have been transferred into the filter. When all the liquid has drained into the flask, rinse the sides of the funnel with acetone to collect all the fibers on the filter paper. Leave the vacuum funnel, then carefully remove the filter paper and fold this over the specimen to insure no loss of material.

Reagents

Sodium hydroxide	400 g solid NaOH dissolved in H_2O and diluted to one liter.
Thiosulfate solution	102 g $Na_2S_2O_3.5\ H_2O$ dissolved in H_2O and diluted to one liter
Sulfuric acid	93-98 percent H_2SO_4, N_2 free
Hydrochloric acid – standard solution	0.1 N
Sodium hydroxide – standard solution	0.1 N
Boric acid solution	40 g H_3BO_4 dissolved in one liter aqueous solution
Potassium hydrogen phthalate	Reagent grade
Phenolphthalein indicator	1 g phenolphthalein dissolved in 200 ml ethyl alcohol
Kjel-Fast catalyst Kjeldahl digestion catalyst	Mercury-free, such as Kelmate NT

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Apparatus. Use a Buchi or CEM Kjel-Fast digester, or equivalent, a pyrex reaction tube or flask, a distillation adapter and a Buchi distillation unit.

Digestion – using the buchi digester. Place the filter paper containing the specimen in a reaction tube and add one packet of Kjeldahl digestion catalyst and 25 ml of concentrated H_2SO_4. Attach the reaction tube to a manifold (by means of a clamp), which in turn is connected to a water aspirator. Set the reaction tube in the Buchi digester, turn on the latter and allow the reaction to proceed for one hour. Turn off the digester and allow the tube and content to cool to room temperature.

Digestion – using the kjel-fast digester. Turn on the CEM Kjel-Fast digestion unit and water supply. (The power switch is on the bottom right corner of the back.. The water spigot is behind the unit.)

The Kjel-Fast unit will prompt:

S-1 SET TIME -

Press 45 ENTER. Then the prompt will be:

S-1 SET POWER –

Press 100 ENTER

There are four stages to the program, the unit will prompt you for the next stage after each is entered. The program is as follows:

S-1 SET TIME = press 45 ENTER
S-1 SET POWER = press 100 ENTER
S-2 SET TIME = press 45 ENTER
S-2 SET POWER = press 100 ENTER
S-3 SET TIME = press 320 ENTER
S-3 SET POWER = press 80 ENTER
S-4 SET TIME = press 100 ENTER
S-4 SET POWER = press 65 ENTER

If a wrong number is pressed, press 0 four times to clear the display. If a wrong number is discovered after ENTER is pressed, press ENTER until the display shows SYSTEM READY then press PROG and start from the beginning. If the program is entered correctly, the system status display will read SYSTEM READY and the time remaining display will read 0740. If

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any problems occur, check the CEM Operation and Service manual for the Kjel-Fast system for more information. With the peristaltic pump in the forward position, lift the switch in the rear of the unit while holding a beaker under the extension tube. Pump Kjel-Fast Catalyst through the system until about 5 mls is pumped through the extension tube. Place the specimen and one packet of Kjeldahl catalyst in the labeled reaction vessel. Add 15 mls concentrated sulfuric acid to the reaction vessel. Place the reaction vessel with insulator on the vessel pedestal inside the microwave cavity. The vessel will catch the Kjel-Fast Catalyst solution that enters the distribution tube. Press start and run the system to completion for about 8 minutes.

Distillation. Before turning on the distillation unit, open the tap water line and check the $Na_2S_2O_3$ solution and the H_2O reservoirs to make sure that these containers have sufficient amounts of these liquids. NOTE: The NaOH-$Na_2S_2O_3$ solution is a mixture of 10N NaOH solution and a 102 g/1 $Na_2S_2O_3 \cdot 5\ H_2O$ in a 5:1 ratio. Turn the power switch on. Install an empty distillation and a receiving vessel into the unit. Preheat the unit for about 5 minutes. Install the distillation vessel containing the test specimen digest and a receiving vessel containing 100 ml of 40 g/1 boric acid solution. Start the distillation and allow this to continue for about 5 minutes. Lower the receiving vessel and allow the collection tube to drip into the vessel. Set the latter aside for titration. Remove the distillation vessel from the unit and pour the remaining liquid into a waste storage container. Rinse the vessel with water and combine the rinsing with the rest in the waste container.

Analysis. Drain the 10-ml buret twice with standardized 0.1N HCL solution. Standardize the pH meter and electrode using a pH4 and a pH7 buffer solution. Titrate the specimen distillate collected from distillation with the standard 0.1N HCL solution to the pH of the blank solution. Note: The blank solution of prepared at least twice and measured for pH each week. Calculate the percent of acrylic fibers in the specimen as follows:

$$\% \text{ Acrylic Fibers} = \frac{6.05 \times V \times N}{W}$$

Where, V = Volume of standard 0.1N HCL solution, in ml.
N = Normality of the HCL solution
W = Weight of test specimens, in grams."

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4.5.1.3.4.1, under Analysis:

Delete: "$\% \text{ Acrylic Fibers} = \frac{6.05 \times V \times N}{W}$

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Where, V = Volume of standard 0.1N HC1 solution, in ml.
N = Normality of the HC1 solution.
W = Weight of test specimens, in grams."

And substitute: "% Acrylic Fibers = $\frac{140 \times V \times N}{KW}$

Where, V = Volume of standard 0.1N HC1 solution, in ml.
N = Actual normality of the HC1 solution.
W = Weight of test specimens, in grams.
K = Nitrogen content of acrylic fibers, in %."

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